

Aging, Selection and Sectoral Productivity Gap

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[Latest Version]

[Online Appendix]

Abstract

This paper studies how population aging affects sectoral productivity through labor reallocation and selection. I document that countries with older populations tend to have larger service sectors and lower service–manufacturing productivity gaps. To explain this pattern, I develop a quantitative Roy model embedded in an overlapping-generations framework with non-homothetic preferences. As retirees exhibit stronger demand for services, aging raises the relative price of services and relaxes the sectoral selection threshold. Workers sort based on comparative advantage, so marginal entrants into services are less productive, reducing average service-sector productivity. Calibrated to U.S. micro data and evaluated using cross-country panel data from 2000–2017, the model explains 36% of the decline in the U.S. productivity gap and closely matches service employment dynamics across high-income countries. Extending the framework to allow for skill-biased and capital-augmenting technological change, I find that selection remains quantitatively important. The results suggest that both demand-side structural change driven by population aging and production-side structural change induced by biased technological change operate through a common general-equilibrium selection mechanism. In both cases, sectoral expansion reallocates workers and makes selection quantitatively important for explaining movements in the service–manufacturing productivity gap.

1 Introduction

Among the rich literature on structural transformation, Cravino et al. (2022) document that population aging has contributed to the rise of service economies in the United States. This naturally raises the question of whether similar patterns hold across countries. Does population aging universally stimulate service demand and expansion? If so, what are the implications – if any – for sectoral productivity levels? In this paper, I document that for many countries, productivity levels are higher in the service sector than in the manufacturing sector. In countries with larger service sectors, however, productivity levels in services fall, relative to manufacturing productivity. Thus, aging is negatively correlated with the service–manufacturing productivity ratio. Countries with older populations have larger service sectors, and their service sectors are relatively less productive than we observe in countries with younger populations. Figure 1 documents the relationship between country demographic profiles and the ratio of productivity between services and manufacturing. As is evident, in all four countries, population aging is associated with a decline in the ratio of service productivity to manufacturing productivity. (Baumol, 1967).

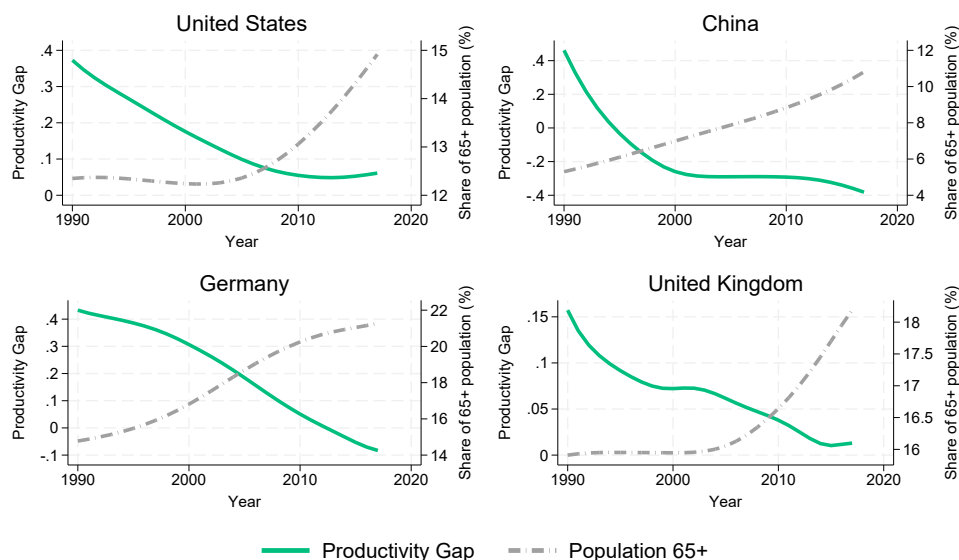


Figure 1: Cross-Country Comparison: Aging versus Productivity Gap

Note: The green solid line plots the service–manufacturing productivity gap for countries in the Panel over 1990–2017 (left y-axis), while the gray dashed line shows the share of population aged 65 and above over the same period (right y-axis). The productivity gap is calculated as service-sector productivity minus manufacturing productivity (PPP adjusted). For category definitions see Online Appendix, and the calculation method for productivity gap follows Gollin et al. (2014).

Although multiple interpretations of the data are possible, in this paper, I draw on a framework

from Lagakos and Waugh (2013) to understand the patterns of growth and sectoral productivity. This paper in turn makes use of Roy (1951)’s selection model. In this framework, as the service sector expands, it draws new workers from other industries through self-selection. Under some assumptions about the joint distribution of skills, these entrants possess lower sector-specific efficiency in services than incumbent workers. This compositional shift leads to a decline in measured productivity growth per worker, even though productivity per efficiency unit of labor remains constant.

This paper makes three main contributions. First, it provides a demographic-based explanation for cross-country differences in sectoral productivity levels, linking population aging to the service–manufacturing productivity gap through a general-equilibrium selection mechanism. Second, it embeds Roy-style sectoral selection into an overlapping-generations (OLG) model with non-homothetic preferences, allowing demographic transitions to generate persistent and quantitatively meaningful productivity dynamics. Third, it shows that even when biased technological change—through skill-biased and capital-augmenting channels—is incorporated, selection effects remain quantitatively important in explaining productivity-gap changes in the United States.

This paper also contributes to a broader literature on the rise of service-based economies and the potential macroeconomic consequences of service-sector expansion (Duarte and Restuccia, 2010; Buera and Kaboski, 2012; Comin et al., 2021). Other work makes use of different mechanisms to explain the growth of services. For instance, Ngai and Pissarides (2007) rely on a model in which productivity growth exogenously differs between manufacturing and services. Under the assumption that services and manufactured goods are gross complements, faster productivity growth in manufacturing leads to a version of Baumol’s cost disease, in which the service sector draws in productive resources (Ngai and Pissarides, 2007; Young, 2014; Duernecker et al., 2024). While these technology-driven explanations are important, this paper highlights a distinct demographic channel: as societies age, demand shifts toward services, raising their share and relative price. The higher relative price relaxes the selection threshold, drawing more workers into services. Because entry is based on comparative rather than absolute advantage, average service-sector productivity declines as composition shifts toward lower-ability workers. In practice, demographic and technological forces may have an comprehensive effect on selection, and the extended model below allows for both mechanisms.

Under the general-equilibrium Roy framework with non-homothetic preferences, the benchmark model yields two key propositions linking demographic aging to sectoral outcomes. First, as retirees have stronger preferences for service goods, population aging raises aggregate service demand, and thus increases the relative price of services. Second, a higher relative price relaxes the self-selection threshold, drawing in workers with weaker comparative advantage in services and thereby lowering average service-sector productivity. Together, these results imply that aging economies experience higher service prices but slower service productivity—providing a theoretical foundation for the relatively lower productivity growth of the service sector observed across

countries.

Using the benchmark calibration and U.S. data from 2000 to 2017, the model reproduces 36% of the observed change in the productivity gap, 89% of the increase in the service-sector value added share, and 57% of the rise in the service-sector employment share. Extending the same calibration to a broader cross-country panel, the model continues to perform reasonably well in predicting levels of sectoral outcomes in recent years. In particular, predictions for the employment share of the service sector are especially strong, with a cross-country coefficient close to one (0.99), indicating that the model nearly matches observed cross-country variation in levels. In terms of dynamics, performance varies across outcomes. The model captures changes in the productivity gap primarily among developed countries, explaining 17% of the observed variation. However, because the benchmark specification abstracts from sector-specific technological change, it tends to underestimate the magnitude of productivity-gap dynamics ($\beta = 4.44$). Predictions for value added share changes are less accurate across countries. By contrast, the model performs best in predicting employment share dynamics: among developed countries, the estimated slope coefficient is 1.03, and the model explains 13% of the variation, indicating that the model closely tracks observed employment reallocation patterns.

The findings of my benchmark model are consistent with studies that employ parametric skill distributions, showing that productivity growth in an expanding sector tends to decline as new, less efficient workers enter (Lagakos and Waugh, 2013; Adão, 2015; Alvarez-Cuadrado et al., 2025). Results from the benchmark analysis show that aging alone has strong explanatory power for structural transformation and productivity-gap dynamics in the United States and other highly aged economies, but performs less well in many developing countries where structural differences and technological heterogeneity are more pronounced.

To isolate the pure selection effects driven by demand-side changes, the benchmark model abstracts from sectoral production heterogeneity and exogenous technological change. While this setup clarifies the demand-driven mechanism, it limits the extent to which the model can account for the observed changes in the productivity gap. These limitations motivate an extended model that incorporates biased-technological change. The extended model provides more margin for the selection mechanism compared to the demand-driven benchmark model. In particular, the extended model is characterized by skill-biased technical change (SBTC) and capital-augmenting technical change (CATC), following the parameterization and empirical evidence in Buera et al. (2022) and Hubmer (2023), based on BEA-KLEMS and World-KLEMS data. Allowing for exogenous technological change and capital-labor substitution expands the production-side margins of adjustment. Nevertheless, I find that selection effects continue to play a quantitatively important role in explaining the change in the productivity gap in the United States.

Related Literature

A large literature in macro-development documents sizable productivity gaps across sectors and highlights frictions in labor reallocation. Early work focuses on agriculture, showing that barriers to input adoption and mobility generate persistent agricultural productivity gaps (Restuccia et al., 2008; Adamopoulos and Restuccia, 2014; Gollin et al., 2014). Selection plays a central role: productivity differences partly reflect worker sorting across sectors, as shown by Gai et al. (2021); Cao et al. (2025) for China and Lagakos and Waugh (2013); Alvarez-Cuadrado et al. (2025) for the cross-country agricultural gap. These papers emphasize that misallocation and selection jointly shape sectoral productivity, providing a foundation for our focus on labor reallocation between services and manufacturing. Recent work turns to the service sector’s rise and its distributional consequences. Leon-Ledesma and Moro (2020) link investment-specific technological change to service expansion, while Fang and Herrendorf (2021); Fan et al. (2023) show heterogeneous welfare effects of service-led growth in China and India. Service-sector growth also interacts with gender gaps: Ngai and Petrongolo (2017) show that expanding services increase female employment, and related Roy-model studies highlight frictions shaping sectoral entry (Lee, 2024; Cao et al., 2025). More broadly, sectoral reallocations contribute to aggregate productivity dynamics (Duarte and Restuccia, 2010), and micro evidence confirms selection into services based on comparative advantage (Shu, 2025). We contribute to this literature by showing how demographic shifts alter the composition of workers and reallocation patterns, generating service productivity declines.

Our second strand is demographic change. Classical overlapping-generations frameworks (Yaari, 1965; Blanchard, 1985; Aiyagari, 1994; Gertler, 1999) map changing age structure to savings, labor supply, and equilibrium outcomes. Empirical work documents multiple channels linking aging to growth: changing labor supply, capital deepening, and technology responses (Cutler et al., 1990; Eggertsson et al., 2019; Acemoglu and Restrepo, 2022). While some argue aging primarily affects labor supply, recent evidence highlights productivity effects as well. Aging alters managerial turnover and entrepreneurship (Liang et al., 2018), dampens competition through consumer inertia (Bornstein, 2025), and reduces regional GDP growth through slower employment and productivity growth (Maestas et al., 2023). We add to this literature by focusing on a specific mechanism: aging reshapes the skill and age composition of labor across sectors and reduces aggregate productivity through reduced sectoral productivity.

2 Motivating Facts

2.1 Cross Country Evidence

In the cross country analysis, we use country-level macro data to illustrate a descriptive relationship of aging on productivity gap of service and manufacturing sector. The data used are mainly drawn from the World Bank Global Productivity data, WDI and Penn World Table. The World

Bank Productivity Project documents 9 sectoral and aggregated value added, employment and productivity statistics for 103 countries up to 2017. The classification of the production sectors follows a constructed category in Online Appendix. To make the productivity data comparable, we use 2010 international US dollar to obtain PPP adjusted value added and productivity for subsequent analysis. The aggregated sectoral productivity is calculated by

$$Prod_{i,t}^s = \frac{\sum_{j=g}^{G^s} Value_{i,t}^s / PPP_{i,t}^{2010}}{\sum_{j=g}^{G^s} Emp_{i,t}^s}, \quad s = \{\text{Service, Manufacturing}\},$$

where i is country, t is year and G is the classified groups. The calculation of productivity gaps follows Gollin et al. (2014), is given by

$$prodgap_{i,t} = \frac{Prod_{i,t}^S}{Prod_{i,t}^M} - 1.$$

I construct a cross-country panel dataset that allows us to conduct descriptive regression analysis. By examining the two-dimensional data on aging levels and productivity gaps, we can easily observe their correlation. However, in a panel setting, cross-country heterogeneity makes it difficult to identify this relationship from simple correlations alone, since the level of calculated productivity level varies significantly across countries. The reduced form regression is given by

$$prodgap_{i,t} = \beta_0 + \beta_1 65share + X'_{i,t} \gamma + \mu_i + \lambda_t + \epsilon_{i,t}, \quad (1)$$

where $X_{i,t}$ is a vector of covariates¹ which will control for country level heterogeneity in development level, standard errors are clustered at the country level. The estimated coefficient of -0.043 in Table 2 Panel (2a) suggests that a one percentage point increase in the share of population aged 65 and above is associated with a 0.043 reduction in the productivity gap. The calculated elasticity for aging is $\eta = -15.72\%$, which means that the partial effect of share of population aged 65 and above is economically significant.

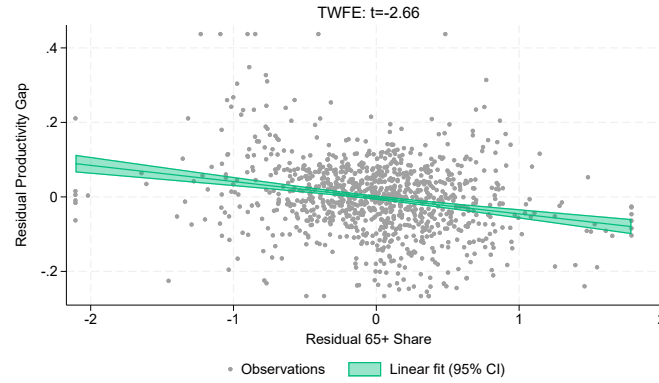
I also want to re-examine whether aging will lead to the expansion of the service sector as one of the drivers of structural transformation as Cravino et al. (2022) have shown. The reason of the re-examination is that we want to use aging and the expansion of the service sector as one of the mechanisms to show the productivity shifts. I use the same covariates in (1) to control for country level heterogeneity, and standard errors are clustered at the country level. The reduced form is

¹These covariates include: (1) Real GDP per capita, (2) share of government expenditure in GDP, (3) share of aggregate saving, (4) share of international trade; (5) person engaged in employment, (6) average annual hours worked by persons engaged, (7) human capital index, (8) capital service, (9) Real capital stock at constant national prices, (10) share of labour compensation in GDP, (11) population. (1)-(4) are drawn from World Bank database and (5)-(11) are drawn from PWT.

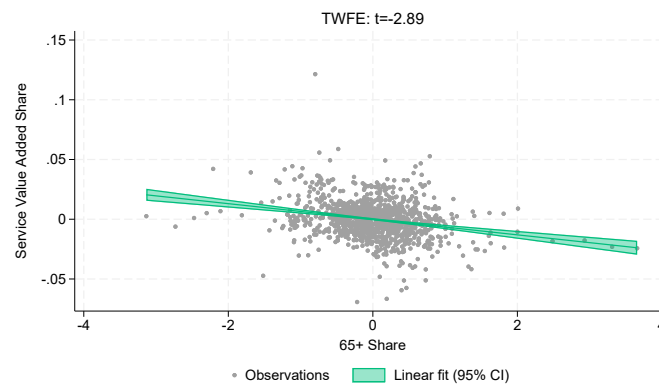
given by

$$servshare_{i,t} = \beta_0 + \beta_1 65share + X'_{i,t} \gamma + \mu_i + \lambda_t + \epsilon_{i,t}, \quad (2)$$

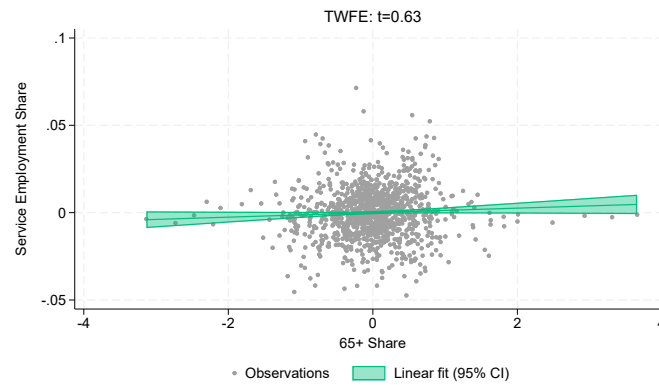
where $servshare_{i,t}$ denotes the share of value added in the service sector relative to total value added or the share of employment in the service sector of country i at year t . Panel (2b) of figure 2 shows that, after controlling for two-way fixed effects, the coefficient on aging for the service value-added share remains statistically significant, but becomes negative. Panel (2c) further indicates that, under the same specification, aging's effect on the share of service employment is not statistically significant. Once fixed time effects are introduced, these estimates run counter to the empirical evidence documented in Cravino et al. (2022), who find that aging is associated with an expansion of the service sector. In short, the reduced-form evidence points to the need for structure: the model allows us to cleanly isolate the contribution of demographic change and trace its general-equilibrium implications for sectoral allocation and productivity.



(a) Productivity Gap



(b) Value Added Share



(c) Employment Share

Figure 2: Partial Correlation of Aging on Variables

Notes: Each gray dot in the plots is a country-year observation. On the y-axis is the residual of a regression of the service-manufacturing productivity gap (Panel a), Value-Added share of service (Panel b), and employment share of service (bottom Panel) on real GDP per capita, and other variables represent economic development level.

2.2 Consumer Expenditure Evidence

In this section, we use consumer-level microdata to support a key assumption of the theoretical model—that older individuals have a stronger preference for services. The analysis draws on Public-Use Microdata (PUMD) from the Interview Survey of the Bureau of Labor Statistics’ (BLS) Consumer Expenditure Survey (CEX). This survey spans more than 40 years, from 1982 to 2023, during which approximately 10,000 addresses are contacted each quarter. It provides detailed Panel data on expenditures across more than 350 categories of goods and services, along with rich demographic information (the most recent release being PUMD 2023). This dataset has been widely used in studies of structural transformation and consumer behavior, including Boppart (2014), Aguiar and Bils (2015), Comin et al. (2021), Cravino et al. (2022), and among others. The main survey files used in this section are: (1) FMLI, which contains consumer-unit-level summary expenditures, income, characteristics, and weights; (2) MTBI, which provides monthly expenditures; and (3) MEMI, which records member-level income.

To categorize consumption, we follow the classification of Aguiar and Bils (2015), who grouped the original 350 CES categories into 24 broad expenditure types, and Cravino et al. (2022), who further divided these broad types into goods and services. Family member records are aggregated to the consumer-unit (CU) level to construct age-structure bins and average age. We exclude households with zero or negative service expenditures and restrict the sample to urban households. I calculate sectoral expenditure shares for four periods: 1982-1991, 1992-1991, 2002-2011, and 2012-2023 using CEX data² Within services, housing expenditures represent a large share. However, we exclude housing in the subsequent analysis, since the rental value of owner-occupied housing is self-reported and may not be directly comparable to rent payments reported by renters in the CEX.

The CEX Interview Survey is retrospective. For example, if the survey is conducted in January 2017, the reported expenditures cover the period from October to December 2016, yielding a month scope of three, since all three months of 2016Q4 are included. By contrast, if the survey is conducted in March 2017, it covers only December 2016, resulting in a month scope of one. To account for representativeness, CEX assigns weight (*finlwt21*) to each consumer unit according to the significance of representative. When aggregate the survey data, the weight applied for computing annual aggregates is given by

$$repwt_{i,t} = \frac{finlwt21_{i,t} \cdot monthscope_{i,t}}{12}, \quad \text{if } finlwt21_{i,t} > 0.$$

I conduct a regression of age dummies on the expenditure share of service, controlling for consumer units, income decile, and region, time effects to capture the relative prices differences and shifts across states and time, following the construction of Aguiar and Bils (2015) and Comin

²The detailed classification is shown in Online Appendix B.

et al. (2021). The reduced form estimation is given by

$$\begin{aligned} expshare_{it} = & \alpha + \sum_{a=1}^{12} \beta_a \mathbf{1}\{agegroup_i = a\} + \gamma_1 famsize_{it} + \gamma_2 earners_{it} \\ & + \gamma_3 income_{it} + \gamma_4 totexp_{it} + \mu_i + \lambda_t + \varepsilon_{it}, \end{aligned} \quad (3)$$

where the age group [30, 34) serves as the baseline; *famsize* denotes family-size dummies; *earners* denotes dummies for the number of earners in the family; *income* denotes income-quartile dummies; *totexp* controls for the total expenditure of the consumer unit; and we also include two-way fixed effects. Figure 3 plots the estimated coefficients and confidential interval bars. The results show a clear positive age gradient in the expenditure share on services: coefficients are close to zero for households younger than 40, increase to about 0.05-0.10 for those in their 50s, and rise further to around 0.15-0.20 for households in their 60s and 70s. The effect is strongest for the oldest group (80 and above), where the coefficient approaches 0.25. The pattern is remarkably consistent across decades, although the estimates in more recent periods tend to be slightly larger.

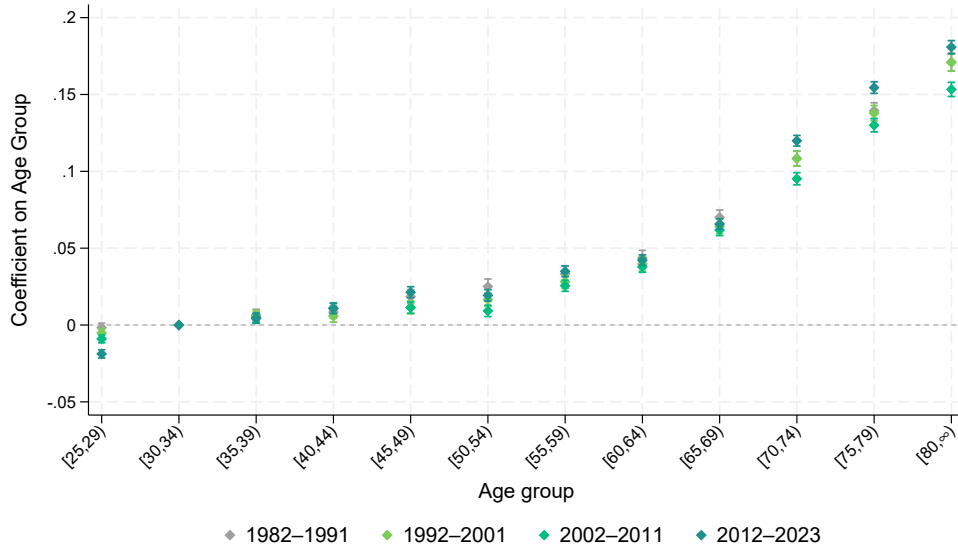


Figure 3: Coefficient Plot by Age Groups and Decades

Note: This figure displays the coefficients of age dummies resulting from estimating equation (3). Each dot represents the point estimate of the age dummies for a particular decade in the CES data. The omitted dummy is that of age group 25-30. The bands report the 95 percent confidence intervals based on standard errors clustered at the household level.

3 Benchmark Model

In this section, we develop a two-sector OLG-Roy model with workers and retirees evolving via constant transition probabilities and fair annuities market following the Yaari-Blanchard

setting, which keeps aggregation and evolution process tractable and controllable. Production in services and manufacturing are simplified by using identical technologies with sectoral effective labor as the key input. Workers are heterogeneous in sector-specific skills, and will self-select themselves across sectors according to their relative advantage decided by their given skill level. The productivity gap between service and manufacturing is driven by aging from the following mechanism: the rising level of aging will rise the relative price for services and lower the relative threshold for workers to self-select into service sector and the process of self-selection is based on the comparative advantage of the worker. We follow Lagakos and Waugh (2013) and assume that a worker's comparative advantage is positively correlated with their absolute advantage. This implies that as a sector expands and labor reallocates, the marginal entrants may be less productive than the incumbents. In the benchmark model, sectoral labor reallocation is driven solely by the selection effect, that is, by workers sorting across sectors according to their comparative advantage in sector-specific skills.

3.1 Evolution of Population

Each individual is born a worker. Conditional on being a worker in the current period, the probability of remaining in one in the next period is ω . Conversely the probability of retiring is $1 - \omega$. Conditional on being retired in the current period, the probability of surviving to the next is γ and, conversely, the probability of death next period is $1 - \gamma$. The population grows at the net rate n . In particular each period $(1 - \frac{\omega}{1+n})N_t$ new workers are born, which follows

$$N_t = (1 + n)N_{t-1}.$$

The total work force at time t is N_t , and the number of retirees is $\left(\frac{1-\omega}{1+n-\gamma}\right) N_t$. The ratio of retirees to workers, ψ is

$$\psi \equiv \frac{1 - \omega}{1 + n - \gamma}$$

Here we assume perfect annuities market following Yaari (1965) and Blanchard (1985). The annuities market provides perfect insurance against this kind of risk. Under the arrangement each retiree effectively turns over his wealth to a mutual fund that invests the proceeds. The fraction γ of those that survive to the next period receive all the returns, while the fraction $1 - \gamma$ who die receive nothing. Each surviving retiree receives a return that is proportionate to his initial contribution of wealth to the mutual fund. Thus, for example, if R is the gross return per dollar invested by the mutual fund, the gross return on wealth for a surviving retiree is R/γ .

3.2 Environment

Retiree's Problem The retiree's utility are given by the following PIGL indirect utility function (Eckert and Peters, 2022; Gollin et al., 2025)

$$V(e_t^r, \mathbf{p}_{s,t}) = \frac{1}{\eta^r} \left(\frac{e_t^r}{\mathbf{p}_{s,t}^{\phi^r}} \right)^{\eta^r} - \nu^r \ln \mathbf{p}_{s,t}, \quad \eta^r, \phi^r \in (0, 1),$$

where manufacturing goods' price is normalized to 1 and $\mathbf{p}_{s,t}$ is service sector's relative price. η^r, ϕ^r, ν^r are consumer type specific parameters, controlling for Engel elasticity, asymptotic share of service expenditure and taste shifter for relative price. Retiree maximize their life time utility subject to budget constraint

$$A_{t+1}^r = (R_t/\gamma)A_t^r - e_t^r.$$

We obtain the Euler equation

$$e_{t+1}^r = e_t^r \left[\frac{\mathbf{p}_{s,t+1}^{\phi^r \eta^r}}{\mathbf{p}_{s,t}^{\phi^r \eta^r}} \cdot \frac{\gamma}{\beta_r R_{t+1}} \right]^{1/(\eta^r - 1)}. \quad (4)$$

The expenditure share of services is given by Roy identity

$$\vartheta_s^r(e_t^r, \mathbf{p}_{s,t}) = \phi^r + \nu^r \left(\frac{\mathbf{p}_{s,t}}{e_t^r} \right)^{\phi^r}. \quad (5)$$

Worker's Problem The workers are indexed by i , who differ in productivity, as explained below. Preferences are given by

$$V(e_{i,t}^w, \mathbf{p}_{s,t}) = \frac{1}{\eta^w} \left(\frac{e_{i,t}^w}{\mathbf{p}_{s,t}^{\phi^w}} \right)^{\eta^w} - \nu^w \ln \mathbf{p}_{s,t}, \quad \eta^w, \phi^w \in (0, 1).$$

Each worker is endowed with one unit of time which she supplies inelastically to the labor market. Each worker is endowed with a vector of individual productivities, denoted $\{z_s^i, z_m^i\}$, which represent the efficiency of one unit of labor in sector s and m . Individual productivities are drawn from a distribution $G(z_s^i, z_m^i)$ with support on the positive reals. The endowed productivities will not change over a individual's life time. The budget constraint of worker i is

$$A_{i,t+1}^w = R_t A_{i,t}^w + W_{it} - e_{i,t}^w$$

Similarly we obtain the Euler equation and expenditure share of services

$$e_{i,t+1}^w = e_{i,t}^w \left[\frac{\mathbf{p}_{s,t+1}^{\phi^w \eta^w}}{\mathbf{p}_{s,t}^{\phi^w \eta^w}} \cdot \frac{\gamma}{\beta_w R_{t+1}} \right]^{1/(\eta^w - 1)} \quad (6)$$

and

$$\vartheta_s^w(e_{i,t}^w, \mathbf{p}_{s,t}) = \phi^w + \nu^w \left(\frac{\mathbf{p}_{s,t}^{\phi^w}}{e_{i,t}^w} \right)^{\eta^w}. \quad (7)$$

Production In the production side, the market in both sectors has its own aggregate production function. To simplify the model we assume the technologies operated by the competitive entrepreneurs in both sectors are identical and freely available. The technologies are given by

$$\mathbf{Y}_{s,t} = X_t^s \cdot L_{s,t} \quad \text{and} \quad \mathbf{Y}_{m,t} = X_t^m \cdot L_{m,t},$$

where X_t is exogenous and captures the level of economy-specified technologies, and $L_{s,t}$ and $L_{m,t}$ represent the total number of effective labor units employed in the two sectors.

Let $\Omega_{s,t}$ $\Omega_{m,t}$ denote the sets of workers choosing to work in service and manufacture, the sector aggregated labor inputs $L_{s,t}$ and $L_{m,t}$ are defined as

$$L_{s,t} \equiv \int_{i \in \Omega_{s,t}} z_s^i dG_i \quad \text{and} \quad L_{m,t} \equiv \int_{i \in \Omega_{m,t}} z_m^i dG_i$$

which represents the sum of all individual productivity employed in two sectors. The total number of workers in each sector is defined as

$$N_{s,t} \equiv \int_{i \in \Omega_{s,t}} dG_i \quad \text{and} \quad N_{m,t} \equiv \int_{i \in \Omega_{m,t}} dG_i.$$

3.3 Optimization and Equilibrium

An equilibrium of the economy consists of a relative service price $\mathbf{p}_{s,t}$, wages per efficiency unit of labor in each sector and allocations for all workers, where workers optimize and both labor markets and output markets clear. The measured labor productivity in the equilibrium is calculated by $\mathbf{Y}_{s,t}/N_{s,t}$ and $\mathbf{Y}_{m,t}/N_{m,t}$, which is the physical quantity of output produced per worker in each sector. Workers will first choose which sector to supply her labor, and maximize her utility subject to the budget constraint, and in competition markets, the wages per efficiency unit of labor are defined as

$$w_{s,t} = \mathbf{p}_{s,t} X_t \quad \text{and} \quad w_{m,t} = X_t.$$

Working in service sector is optimal for worker i if and only if

$$W_{s,t}^i \geq W_{m,t}^i \Rightarrow z_s^i \mathbf{p}_{s,t} X_t \geq z_m^i X_t.$$

Thus we obtain the simple cutoff rule (8). With sufficient high relative productivity in service sector, the worker will optimize her labor income by entering the service sector. With higher $\mathbf{p}_{s,t}$, the relative threshold for workers to self-select into service sector will decrease,

$$\frac{z_m^i}{z_s^i} \leq \mathbf{p}_{s,t}. \quad (8)$$

Using this cutoff rule, we will obtain the sets of employed workforce $\Omega_{s,t}$ and $\Omega_{m,t}$ at equilibrium:

$$\Omega_{s,t} = \left\{ i \mid \frac{z_m^i}{z_s^i} \leq \mathbf{p}_{s,t} \right\}, \quad \Omega_{m,t} = \left\{ i \mid \frac{z_m^i}{z_s^i} > \mathbf{p}_{s,t} \right\}. \quad (9)$$

Because the model features exogenous population growth, we focus on a per-capita steady state equilibrium, in which variables expressed in per-capita terms remain constant over time. Define per-capita wealth of retirees $a_t^r \equiv \frac{\mathbf{A}_t^r}{\psi N_t}$, where $\mathbf{A}_t$ is the aggregate wealth at t , and the aggregate wealth accumulation process of retirees is given by

$$\underbrace{\psi N_{t+1} a_{t+1}^r}_{\mathbf{A}_{t+1}^r} = R_t \underbrace{\psi N_t a_t^r}_{\mathbf{A}_t^r} + (1 - \omega) N_t \mathbb{E}[\mathbf{A}_{t+1}^w], \quad (10)$$

dividing ψN_t , we obtain the per-capita wealth accumulation equation of retirees

$$a_{t+1}^r = \frac{R_t}{1+n} a_t^r + \frac{1-\omega}{1+n} \mathbb{E}[\mathbf{A}_{t+1}^w]. \quad (11)$$

The aggregate wealth accumulation of workers is given by

$$\underbrace{N_{t+1} a_{t+1}^w}_{\mathbf{A}_{t+1}^w} = \omega \left(R_t \underbrace{N_t a_t^w}_{\mathbf{A}_t^w} + \underbrace{p_{s,t} X_t L_{s,t} + X_t L_{m,t}}_{\mathbf{W}_t} - N_t \mathbb{E}[e_{i,t}^w] \right), \quad (12)$$

similarly dividing by N_t , the per-capita wealth accumulation equation of retirees is characterized by

$$a_{t+1}^w = \frac{\omega}{1+n} (R_t a_t^w + \bar{w}_t - \mathbb{E}[e_{i,t}^w]), \quad (13)$$

where $\bar{w}_t \equiv \frac{p_{s,t} X_t L_{s,t} + X_t L_{m,t}}{N_t}$. The aggregate wealth of the economy is a simple aggregation of $\mathbf{A}_t^r$ and $\mathbf{A}_t^w$:

$$\mathbf{A}_t = N_t (\psi a_t^r + a_t^w),$$

dividing by the total population of the economy $P_t \equiv (1 + \psi)N_t$, we have $a_t = \frac{\psi a_t^r + a_t^w}{1 + \psi}$. And the accumulation equation follows

$$\underbrace{P_{t+1}a_{t+1}}_{A_{t+1}} = R_t P_t a_t + \underbrace{N_t \bar{w}_t}_{W_t} - (\psi N_t e_{i,t}^r + N_t \mathbb{E}[e_{i,t}^w]), \quad (14)$$

dividing by the total population, we have the per-capita wealth accumulation equation

$$a_{t+1} = \frac{R_t a_t}{1 + n} + \frac{\bar{w}_t}{(1 + n)(1 + \psi)} - \frac{\psi e_{i,t}^r + \mathbb{E}[e_{i,t}^w]}{(1 + n)(1 + \psi)}. \quad (15)$$

3.4 The Effect of Aging

Proposition 1: Consider a economy align with our assumption, if retiree have higher preference on services than workers, population aging will raise the relative price for service goods in equilibrium, i.e. $\partial p_{s,t} / \partial \psi > 0$.³ By our assumption, retirees have a higher subsistence requirement for service goods, , population aging (i.e. a increase in ψ) increases aggregate service good consumption . If the supply of goods is relative inelastic, this rising demand exerts upward pressure on the relative price, potentially explaining structural changes in sectoral prices overtime. We can rewrite the aggregated total consumption as

$$C_s(p_s, \psi) = \psi N \cdot c_s^r + N \int c_{i,s}^w dG = \psi N \cdot c_s^r + N \mathbb{E}[c_s^w]$$

and

$$C_m(p_s, \psi) = \psi N \cdot c_m^r + N \int c_{i,m}^w dG = \psi N \cdot c_m^r + N \mathbb{E}[c_m^w]$$

From the market clearing condition $Y_s = XL_s$, $Y_m = XL_m$, thus at equilibrium, we define

$$D(p_s, \psi) = \frac{C_s(p_s, \psi)}{C_m(p_s, \psi)} - \frac{L_s(p_s)}{L_m(p_s)} = 0.$$

By using implicit function theorem,

$$\frac{dp_s}{d\psi} = -\frac{D_\psi}{D_p} > 0, \quad \text{iff } \vartheta_s^r > \vartheta_s^w,$$

and population aging will raise the relative price for service goods in equilibrium if and only if the expenditure share of service of the retiree is higher than the worker's.

Proposition 2: Consider two identical economies that differ only in aging level ψ . Let $E[z_s | \frac{z_m}{z_s} \leq p_{s,t}]$ be monotone decreasing in $p_{s,t}$. In equilibrium, economy with higher ag-

³Proof in Online Appendix A

ing level will lead to lower service sector productivity, i.e. if $\psi^\alpha > \psi^\beta$, then $\frac{Y_{s,t}^\alpha}{N_{s,t}^\alpha} < \frac{Y_{s,t}^\beta}{N_{s,t}^\beta}$.

Intuitively, as the level of aging increases, by our assumption, the economy's aggregated demand for service goods will increase as a result, so the relative price of service goods in equilibrium will also increase as a response. Consequently, the relative wage in the service sector induces more workers to enter service sector. However, the result relies on a key assumption that the relative advantage of a worker is positively related to its absolute advantage.

In **Proposition 1**, an economy with a higher level of aging will lead to a higher relative price of service goods. From the cutoff rule (8), a higher relative price of service goods will lead to a lower threshold for workers to self-select into the service sector. Consequently, the average individual productivity in the service sector, $E \left[z_s | \frac{z_m}{z_s} \leq \mathbf{p}_{s,t} \right]$, is decreasing $\mathbf{p}_{s,t}$. So, as ψ increases, the level of aging increases leading to higher $\mathbf{p}_{s,t}$, the new threshold for self-selection will lead to a more convenient condition for workers to have comparative advantage when working in the service sector. Thus, we will have

$$\text{if } \psi^\alpha > \psi^\beta, \quad \text{then} \quad \frac{Y_{s,t}^\alpha}{N_{s,t}^\alpha} < \frac{Y_{s,t}^\beta}{N_{s,t}^\beta}.$$

3.5 Skill Distribution

In this section, we will discuss and construct a appropriate skill joint distribution $G(z_s, z_m)$ for our baseline model. To make **Proposition 2** hold, we must have $E \left[z_s | \frac{z_m}{z_s} \leq \mathbf{p}_{s,t} \right]$ monotonically decrease in $\mathbf{p}_{s,t}$. This property on the conditional expectation can be addressed by imposing correlation between the sector specific skills. In the quantitative analysis of Lagakos and Waugh (2013), skills are positively correlated, and the worker who is relatively more productive is also relatively more productive in the other sector. This imposed correlation allows flexibility in matching the data. However, as shown in Shu (2025) and Alvarez-Cuadrado et al. (2025), the micro-level selection pattern suggests that the correlation between service and manufacturing skills may not be positive. We follow the construction of Shu (2025) to qualitatively assess what is the sufficient assumption to support **Proposition 2**.

Assume skills are independently drawn from a Fréchet distribution, the comparative advantage of workers is given by

$$\kappa(q) = \kappa \ln \frac{q}{1-q}, \quad (16)$$

where q is the percentile of the worker's comparative advantage. The absolute advantage of the worker is given by

$$\alpha(q) = \bar{\alpha} + \alpha \ln q. \quad (17)$$

And the sector specific skills are now a composite of comparative and absolute advantage given its percentile. We let the skills of service to be the combination of comparative and

absolute advantages since in our sample, most countries start with positive service-manufacturing productivity gap.

$$\ln z_s(q) = \alpha(q) + \kappa(q), \quad \text{and} \quad \ln z_m(q) = \alpha(q).$$

If $\alpha > 0$, then we will have

$$\ln z_s(q) = \bar{\alpha} + (\alpha + \kappa) \ln q - \kappa \ln(1 - q), \quad \ln z_m(q) = \bar{\alpha} + \alpha \ln q.$$

Since $\alpha > 0, \kappa > 0$, sorting is monotone

$$\ln z_m(q) - \ln z_s(q) = \kappa \ln \left(\frac{1 - q}{q} \right) > 0,$$

in this case as q increases, a worker will have higher comparative advantage in manufacturing. Since a quartile of allocation cutoff, q_s exists when workers are indifferent between the service and manufacturing sector. From the cutoff rule (8), the indifference conditions pins down a q_s such that

$$\ln z_m(q_s) - \ln z_s(q_s) = \ln \mathbf{p}_{s,t}.$$

Solving for q_s , we obtain the cutoff percentile of comparative advantage.

$$q_s(\mathbf{p}_{s,t}) = \frac{1}{1 + \mathbf{p}_{s,t}^{\frac{1}{\kappa}}}, \quad \frac{\partial q_s(\mathbf{p}_{s,t})}{\partial \mathbf{p}_{s,t}} < 0. \quad (18)$$

Following previous derivation, the conditional expectations are given by

$$E[\ln z_s | q \leq q_s] = \bar{\alpha} - \alpha + (\alpha + \kappa) \ln q_s + \kappa \frac{1 - q_s}{q_s} \ln(1 - q_s) \quad (19)$$

From equation (19), we have $\partial E[\ln z_s | q \leq q_s(\mathbf{p}_{s,t})] / \partial q_s(\mathbf{p}_{s,t}) > 0$. Thus the conditional expectations of service skill is monotone increasing in q_s . Combine with equation (18), the conditional expectation of skill level in the service sector $E[\ln z_s | q \leq q_s(\mathbf{p}_{s,t})]$ is monotone decreasing in $\mathbf{p}_{s,t}$. As a conclusion, to ensure **Proposition 2** holds in our environment, it is sufficient that good workers need to stay good even after reallocation, i.e. $\alpha > 0, \kappa > 0$.

Dependent Skill Distribution I follow the construction of Lagakos and Waugh (2013) by using joint Fréchet distribution. Compared to log-normal distributions, Fréchet distribution is more tractable in multi-sector selection problems and the extreme-value distribution allows for fat tails that align with real world skill distribution.

Let skills $\{z_s, z_m\}$ be drawn from a joint distribution

$$F(z_s, z_m) = C[F(z_s), F(z_m)],$$

where the marginal distributions are Fréchet distribution

$$F(z_s) = e^{-z_s^{-\theta_s}} \quad \text{and} \quad F(z_m) = e^{-z_m^{-\theta_m}},$$

and the marginal distributions are linked by a Frank copula

$$C[u, v] = -\frac{1}{\rho} \ln \left[1 + \frac{(e^{-\rho u} - 1)(e^{-\rho v} - 1)}{e^{-\rho} - 1} \right].$$

The key advantage is the ability to model both positive and negative dependence through the parameter ρ . $\rho > 0$ implies positive correlation and $\rho < 0$ implies negative correlation, and $\rho = 0$ implies independence. In the context of cross-sectional skills, this copula allows the model to incorporate different degrees and signs of the correlation between the skills of service and manufacturing while maintaining Fréchet marginal distributions.

4 parameterization

The parameterization in my model follows Table 5.

Table 1: Calibrated Parameters

Parameters	Definition	Value
θ_s	The dispersion level of worker's skill in service	2.638
θ_m	The dispersion level of worker's skill in manufacturing	3.505
ρ	The Frank Coupula correlation	3.762
ν^w	Worker's taste shifter	-0.030
ν^r	Retiree's taste shifter	-0.057
ϕ^w	Worker's asymptotic expenditure share of services	0.504
ϕ^r	Retiree's asymptotic expenditure share of services	0.656
η^w	Worker's Engel elasticity	0.860
η^r	Retiree's Engel elasticity	0.432

4.1 Productivity Distribution

In order to recover the distribution of individual productivity draws, we use data from the U.S. Current Population Survey (CPS), harmonized by CEPR, to calibrate the moments that discipline our Fréchet distribution and Frank copula parameters. Our calibration sample spans the years 2003–2018, during which the CPS classifies employment sectors using the 2-digit NAICS 814 system. We group occupations into manufacturing and services following the classification in

previous section⁴, while excluding workers in agriculture, the armed forces, and those who report never having worked. To ensure consistency, we restrict the sample to individuals whose hourly wages exceed the federal minimum wage and who can be observed in consecutive survey years. Wages are measured as salary income plus $0.66 \times$ business income and $0.44 \times$ farm income, with the income shares following Valentinyi and Herrendorf (2008).

To match the dispersion of wages (used as a proxy for underlying skills), we focus only on the non-transitory component of wage variation. For this reason, I restrict the CPS sample to workers observed in consecutive years, which allows us to separate uncorrelated transitory shocks from the total variance. I estimate a random-effects model to decompose log wages:

$$\ln wage_{i,t} = \mu + \underbrace{\alpha_i}_{\text{Random Effect}} + \epsilon_{i,t},$$

where $Var(\alpha_i)$ captures the variance of the non-transitory component. Using the calculated moments, the calibrated matched dispersion levels of the Fréchet distribution are $\theta_s = 2.638$ and $\theta_m = 3.505$ respectively. I use the difference in the P90–P50 wage gaps between sectors instead of the average wage ratio to identify the dependence parameter ρ in the Frank copula. A higher ρ implies that high-ability workers are more likely to cluster in one sector, raising the upper-tail inequality (P90–P50) in that sector relative to the other. Using the percentile gaps, the copula dependence matched is $\rho = 3.762$.

4.2 Preferences Parameters

To calibrate the parameters of the PIGL utility function, we match the empirical expenditure shares of services across income bins and their overall mean to the corresponding model-implied moments. For retirees, the estimated parameters are $\hat{\phi}^r = 0.656$, $\hat{\nu}^r = -0.038$ and $\hat{\eta}^r = 0.423$; for workers, the estimated parameters are $\hat{\phi}^w = 0.504$, $\hat{\nu}^w = -0.006$ and $\hat{\eta}^w = 0.86$. Figure 4 plots the Engel curve given the calibrated parameters. The higher value of ϕ for retirees indicates a larger asymptotic share of services in their consumption, while a lower η suggests a more concave Engel curve, consistent with stronger but saturated service demand relative to workers. Individual's income is standardized in our model, so the estimated ν is around zero, so the deviations from asymptotic shares are weakly corrected. These values of ν is similar to Gollin et al. (2025), which uses the initial U.S. productivity gap as a target. The negative ν implies that services behave as luxuries, with the expenditure share converged to ϕ from below as income increases, consistent with Aguiar and Bils (2015).

⁴The Classification of the Occupations from CPS is shown in Online Appendix B

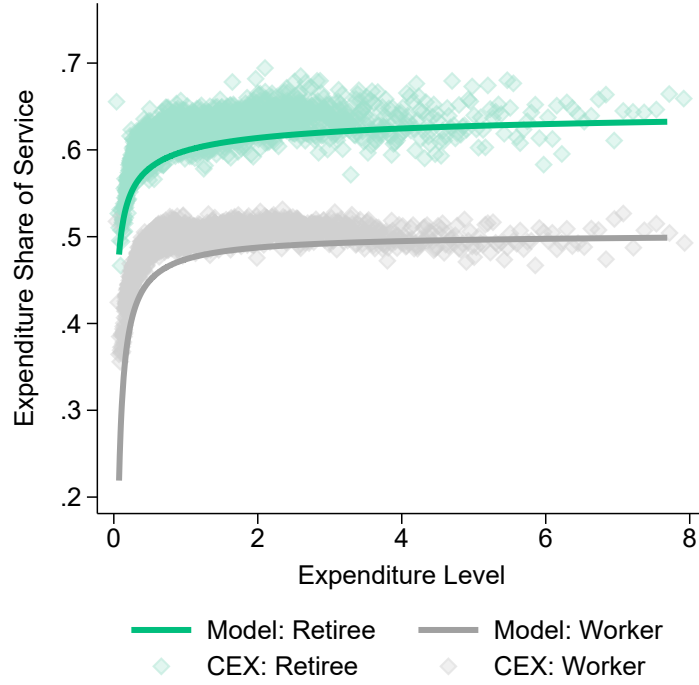


Figure 4: Engel Curves of Retiree and Worker

Notes: The y-axis is the expenditure share of service ϑ_s when the relative price of service is pinned at 1. This figure shows that retiree and worker's expenditure share of service as a function of expenditure level e . The observation from CEX data are clustered into 1000 bins, and individual's expenditure level is adjusted to match the simulated expenditure level

5 Quantitative Analysis

This section examines the comparative static equilibrium for both the United States and across countries. We evaluate how well the model's predictions align with the data, assess its explanatory power, and explore the main mechanisms underlying cross-country differences in productivity gaps.

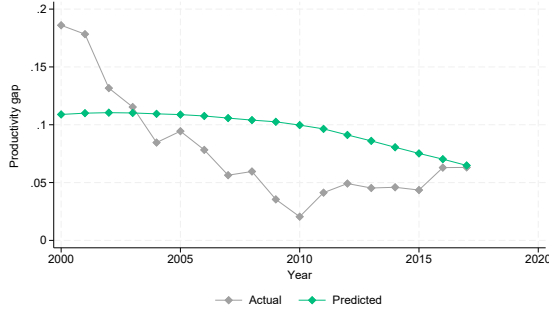
5.1 United States

We first use our benchmark model to predict both the dynamics and levels of United States and some selected countries' outcomes, including value-added shares and employment shares. Given the cross-country heterogeneity in economic characteristics and initial productivity gaps, we proxy the share of retirees ψ in our model using the old-age dependency ratio from the WDI database, and measure the aggregate level of technology by GDP per capita (inflation- and PPP-adjusted) from the Penn World Table. Since the parameterization of the model is tied to United States'

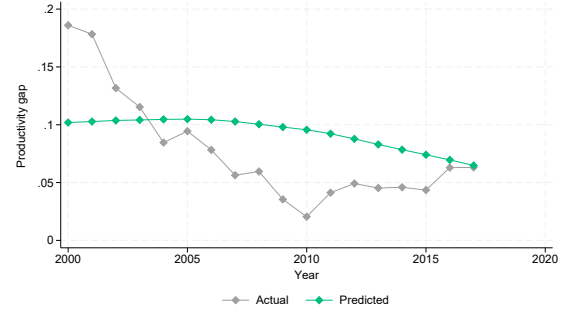
micro-level data, it is more natural to start validating the model from the U.S economy. Rather than imposing the strong requirement that the parameterization and mechanism are suitable across countries.

Figure 5 compares the model's predictions with the actual data for the United States. The observed productivity gap in the United States exhibits much sharper movements: a pronounced compression in the early 2000s, a trough around 2010, and a partial rebound thereafter. These fluctuations reflect shocks that our demographically driven model abstracts from, such as sectoral TFP shocks, cyclical demand fluctuations, and policy-related factors. Similar underestimations occur in other economies that share similar demographic structures and income levels with the United States, suggesting that the impact of demographic selection alone is relatively modest. In Panel (5a), the quantitative dynamics of the model for the United States are driven by both direct income-effect and population aging. Where in Panel (5b), the technology is kept constant. The graphical difference in the predictions is marginal. Table 2 displays the quantitative value of the dynamics of the model and the U.S economy. The value in parentheses is the predicted dynamic value when aging is the only driver of the model, and the quantitative difference between both panels is marginal as well. The predictions are -4.4% and -3.7%, and explain 36% and 30.3% of the observed productivity gap dynamics, respectively. During the sample period, the income per capita of the U.S increased 21.3% and the age dependency of ratio increased 23.5%. Both of the growth are non-negligible, which shows that the Engel Curve for the U.S. economy is relative inelastic in our model. As for other variables, the predicted dynamics of the value-added share in the U.S. economy increased 2.6%, which is very close to the actual dynamics of 2.9%, while the predicted dynamics of the employment share of service is 2.8%, which underestimates the actual dynamics of 4.9%.

Overall, when applying the benchmark model to the U.S. economy, the mechanism of selection and labor reallocation accounts for a substantial share of both the expansion of the service sector and the decline in relative service-sector productivity, even in the absence of exogenous differences in sectoral productivity growth. This finding suggests that, for high-income economies, population aging can exert sizable effects on structural transformation and productivity dynamics through demand-driven reallocation and selection forces, beyond the direct effects of income growth alone.



(a) Income Effects + Aging



(b) Aging Only

Figure 5: Productivity Gap: U.S. Time Series

Notes: This figure compares the model-predicted productivity gaps with the observed productivity gaps for United States from 2000 to 2017. Each gray dot connected by a line represents the realized productivity gap in a given country-year, while each green dot connected by a line represents the corresponding model-predicted value. All reported dynamics are computed as end-of-period values minus beginning-of-period values.

Table 2: Model Dynamics of the United States

	Δ Prod. Gap	Δ VA. Share	Δ Emp. Share
Data	-12.2%	2.9%	4.9%
Prediction	-4.4% (-3.7%)	2.6% (2.2%)	2.8% (2.4%)

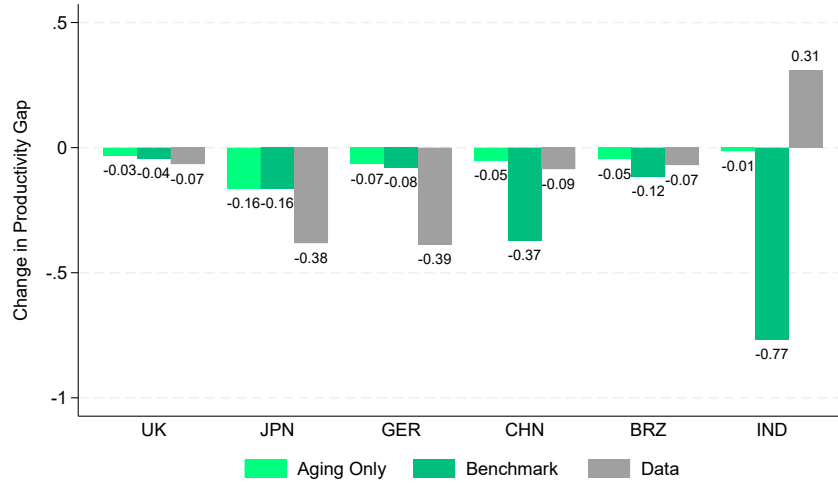
Notes: This table compares the model predicted dynamics of productivity gaps, value-added share of service sector and the employment share of sector from 2000 to 2017. The value in the parenthesis is the model dynamics when technology is constant.

5.2 Selected Countries

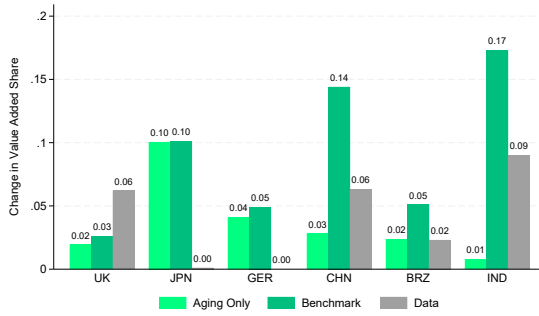
Figure 6 displays the predictions of the model for the dynamics of the value-added share of services and the employment share of services of a set of selected countries. When homogeneous technology progress is imposed across sectoral productions, our model prediction underestimates the observed service-manufacturing productivity gap dynamics, but still explains a substantial fraction as shown in Panel (6a). For lower-income countries, however, the model tends to overstate productivity gap dynamics, as income-driven demand shifts quantitatively dominate the effects of population aging. Figure 6 also shows that the channel of selection and sectoral expansion does not apply to all countries, for example, although the employment share of services in India has increased 7% during the sample period, the productivity of the service sector still increased largely relative to manufacturing. This result is not surprising, as Fan et al. (2023) have shown that in India, the growth in service productivity is the key driver of the expansion of the service sector. Figure 4 provides a potential explanation for these discrepancies: in low-income countries, the

same amount of income growth has a much larger effect on the expenditure share of services than in high-income countries; and the Engel curves for countries like India and China should have different shapes from the one calibrated using U.S. micro-data.

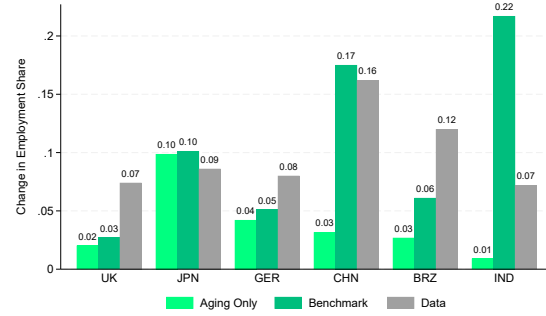
Panel (6b) and (6c) compare the prediction of the model and actual data on the dynamics of value-added share and employment share of service. The selection effects driven by both population aging and income growth generate accurate predictions for the service employment share, although the model tends to overpredict service employment in some developing economies. Relative to its performance in matching employment share dynamics, the model overestimates the increase in the value-added share for most countries in the sample, with the United Kingdom as a notable exception. Importantly, all aggregate outcomes examined in this section are non-targeted moments of the model. Despite this, the model delivers a strong quantitative fit, particularly for countries with income levels close to that of the United States. In these high-income economies, income effects are relatively small, so population aging constitutes the primary force shaping sectoral labor reallocation. As a result, the model tends to underpredict observed service–manufacturing productivity gap dynamics, especially in countries where aging has occurred more slowly than in Japan. Nevertheless, even in these cases, the model explains a substantial share of the observed variation in productivity gaps, highlighting the empirical relevance of the selection mechanism driven by demographic change.



(a) Productivity Gaps



(b) Value-Added Share



(c) Employment Share

Figure 6: Benchmark Model: Selected Countries

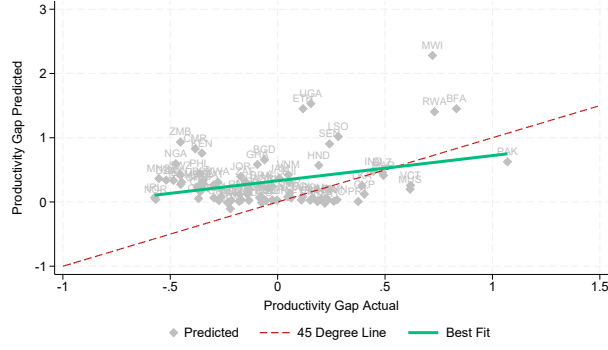
Notes: This graph compares the prediction of the model and the actual data on the dynamics of service-manufacturing Productivity Gaps, the Value-Added Share of service and the Employment Share of service. The light green bars in each panel are the predictions of the model when population aging is the only driver, while the dark green bars are the predictions of the model when both income effects and population aging are considered. The gray bars are the actual data for each selected countries. The selected countries are (in order) the United Kingdoms, Japan, Germany, China, Brazil and India.

5.3 Cross-Country Panel

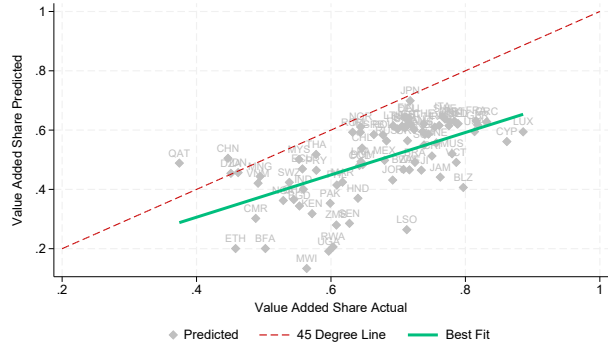
Levels Figure 7 displays the model's predictions compared with actual data. We apply the non-targeted version of the model to all countries in our 2000–2017 sample. The red dashed line in each Panel represents the 45-degree line (perfect prediction), while the green solid line shows the linear best fit between predicted and observed data. The cross-country prediction in Panel (7a) yields a coefficient of 0.39 and a R^2 of 0.10, suggesting that the model captures at least part of the variation in productivity levels. The benchmark model performs better in predicting the expansion of the service sector: Panels (7b) and (7c) show the predictions of the model for the

value-added and employment shares of the service sector, with coefficients of 0.51 and 0.99, and R^2 of 0.36 and 0.73, respectively. These results indicate that, while the selection model accurately predicts labor allocation across sectors, it has a more limited ability to explain differences in output levels. One proposed reason is that the model does not explicitly incorporate sectoral differences in production technologies, so sectoral heterogeneity arises solely from differences in expected skill level of workers. Which leads to our extended model with high-and-low labor inputs and automation choice.

One of our benchmark model's highlights is that the prediction on labor allocation is accurate from 2000 to 2017, which means that the benchmark model successfully captures the relative wage shift and marginal rate of substitution between good and service when using United State's micro-data as calibration targets. Another feature of the benchmark model is that although it correctly predicts a large proportion of variations between countries in value-added shares, it systematically underestimates the level by about 20–30%. This bias likely arises for two reasons. First, the model does not further disaggregate the service sector into its major sub-components, causing it to miss the rapid expansion of certain high-growth service sub-sectors. Second, many service outputs are used as intermediate inputs in production, so treating services purely as final consumption may lead the model to understate their overall demand and contribution to value added. Regarding productivity gaps, the model accurately matches the evolution of the service–manufacturing productivity gap from 2014 to 2017. This pattern suggests that, when using the United States as the benchmark frontier, the selection mechanism operates primarily in the modern period. In earlier decades, the rise of services appears to have been driven less by talent-based selection and more by sector-specific productivity improvements within services themselves.



(a) Productivity Gap: $\beta = 0.39$, $R^2 = 0.10$



(b) Value-Added Share: $\beta = 0.51$, $R^2 = 0.36$



(c) Employment Share: $\beta = 0.99$, $R^2 = 0.73$

Figure 7: Benchmark Model: Prediction vs Actual

Notes: This figure compares the benchmark model predictions with the observed data. Panel (a) plots the predicted versus actual service–manufacturing productivity gap, Panel (b) reports the value-added share in services, and Panel (c) displays the employment share in services. The red dashed line represents the 45-degree line, indicating perfect prediction, while each gray dot denotes the model-predicted value (y-axis) against the observed outcome (x-axis). The green solid line shows the best-fitting linear regression line between predicted and actual values. The comparison is conducted in the last period for the model and the data.

Dynamics Table 3 evaluates the model’s quantitative performance by regressing observed aggregate variables on their model predictions in a cross-country panel. Several patterns emerge. First, for productivity gap dynamics, the model performs poorly in the full sample, yielding a negative slope and low explanatory power. This reflects the dominance of income-driven structural change in lower-income countries, where the productivity growth of service rather than selection effects drives service expansion. In contrast, among top-income countries, the slope rises to 4.44, with an R^2 of 0.17, indicating that the model captures the direction and cross-country variation of productivity gap dynamics but systematically underestimates the magnitude. This underestimation is consistent with muted income effects and slower demographic transitions in many advanced economies relative to the benchmark calibration.

Second, for value-added shares, the model exhibits limited explanatory power in the full sample, with a small positive slope (0.08) and low R^2 (0.03). Among top-income countries, however, the estimated slope turns negative and exceeds unity in absolute value, suggesting that the model overstates value-added share dynamics relative to the data. This discrepancy likely reflects the model’s abstraction from sector-specific price adjustments, measurement error in value-added, and sectoral technology progress that dampen the translation of labor reallocation into value-added shares in practice.

Third, the model performs best in matching employment share dynamics. The estimated slopes are positive and close to unity in both the full sample and the top-income subsample, with R^2 values around 0.12–0.13. This result indicates that the selection and reallocation mechanisms embedded in the model generate accurate predictions for labor reallocation across sectors, even though employment shares are non-targeted outcomes.

It is important to interpret these results in light of the model’s core mechanism. The primary channel through which the model operates is labor reallocation across sectors, driven by population aging and income effects in relative demand. As a result, the model delivers its strongest quantitative performance for employment share dynamics, which are the most direct manifestation of this mechanism. In contrast, value-added shares and productivity gaps are composite outcomes that depend not only on labor or skill allocation, but also on sector-specific productivity shocks, institutional features, and measurement scales. Abstracting from these additional structures will inevitably weaken the model’s ability to match other variables quantitatively. From this perspective, the strong performance of the model in predicting labor reallocation, combined with systematic but interpretable deviations in value-added shares and productivity gaps, suggests that the selection mechanism captures a key component of structural transformation, while leaving room for complementary channels, such as sector-specific technological change or price rigidities, to explain the remaining variation in the observed data.

Table 3: Model Validation: Cross-Country Panel

Outcome	Full Sample (N=82)			Top-30 Income Countries (N=30)		
	Level β	Z-score β	R^2	Level β	Z-score β	R^2
Productivity gap	-0.64	-0.23	0.05	4.44	1.63	0.17
Value-added share	0.08	0.17	0.03	-1.29	-2.58	0.26
Employment share	0.32	0.35	0.12	1.07	1.14	0.13

Notes: Each entry reports the slope coefficient β from a regression of the observed outcome on the model-predicted value, estimated using heteroskedasticity-robust standard errors. Level regressions use variables in levels, while Z-score regressions standardize both observed and predicted series. All coefficients are significant at the 1% significance level. Top-30 Income Countries are defined based on average GDP per capita over 2000–2017.

5.4 Mechanism Analysis

In this subsection, I isolate the demand-side channels through which income effects and population aging influence structural transformation and sectoral productivity. To do so, I conduct a series of counterfactual experiments that separately shut down or activate each mechanism. This decomposition allows me to distinguish the role of income-driven demand shifts from those that arise specifically from demographic change. By comparing these counterfactual results with the benchmark model, I quantify the contribution of population aging to the service–manufacturing productivity gap and examine how its importance varies across countries.

Income Effect I first consider an environment with homogeneous households that share identical non-homothetic preferences, so that income effects constitute the only channel driving structural transformation. Household preferences are still governed by the PIGL utility introduced in Section 4. Calibrating the model to U.S. CEX data, I obtain the parameter values $\phi = 0.517$, $\eta = 0.137$, and $\nu = 0.016$. An important difference relative to the benchmark calibration is that the estimated value of ν becomes positive under homogeneous preferences, whereas it was negative when allowing for preference heterogeneity across workers and retirees. This difference reflects the distinct economic role of the taste shifter in the two environments. One possible interpretation is that because the homogeneous-preference calibration forces a single parameter to match aggregate demand patterns, ν absorbs residual variation previously explained by demographic heterogeneity, leading to a sign change.

Population Aging In order to shut-down the the channel of income effects, I set $\nu = 0$ and the heterogeneous preferences will collapse to Cobb-Douglas with constant expenditure shares. In this case, expenditure shares no longer vary with income, and demand composition is independent of income changes. Calibrating the model to CEX data, we obtain the constant expenditure on

service for retiree and worker, where $\vartheta_s^r = 0.613$ and $\vartheta_s^{rw} = 0.495$. This allows us to isolate the pure demographic channel independent from income effects.

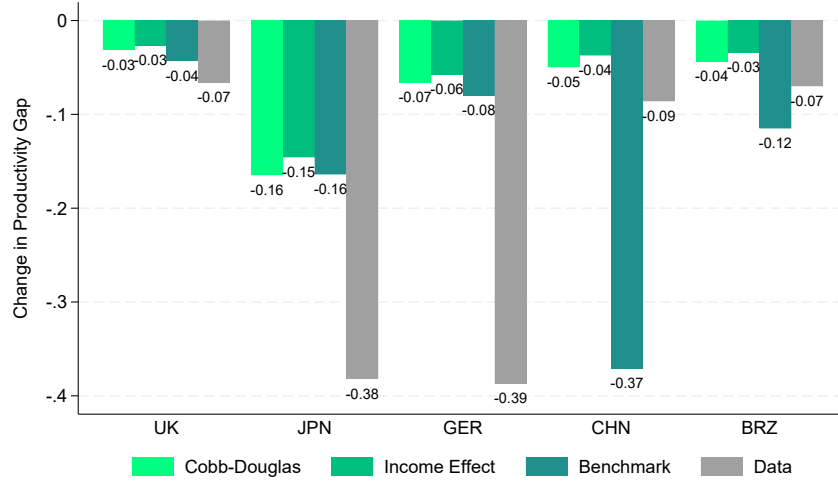
Quantitative Comparison Table 4 and Figure 8 compare the quantitative contributions of income effects and population aging. For the United States, the income effect alone generates a 3.1% decline in the productivity gap, while population aging alone generates a larger decline of 3.8%, accounting for approximately 31% of the observed decline in the data. These results reflect distinct economic mechanisms. Income growth affects demand gradually through non-homothetic preferences, shifting expenditure toward services as households become richer. In contrast, population aging directly increases aggregate service demand through compositional changes, since retirees allocate a larger share of their expenditure to services than workers in our parameterization. As a result, demographic shifts can exert a stronger effect when changes in population composition are substantial relative to income growth.

Compared to the benchmark model, the estimated Engel curve in this counterfactual is flatter, implying a weaker sensitivity of service expenditure shares to income. Consequently, income-driven structural transformation is attenuated, particularly in poorer countries where income growth plays a larger role in driving demand shifts. By contrast, the counterfactual generates predictions that are closer to the benchmark in richer countries, where income effects are more limited and demographic composition plays a more prominent role in shaping aggregate demand. The benchmark model still provides the best prediction on the dynamics of employment share in service. Compared to the counterfactual experiments, the benchmark model captures the selection process with more precision when both income effects and demographics change are well accounted.

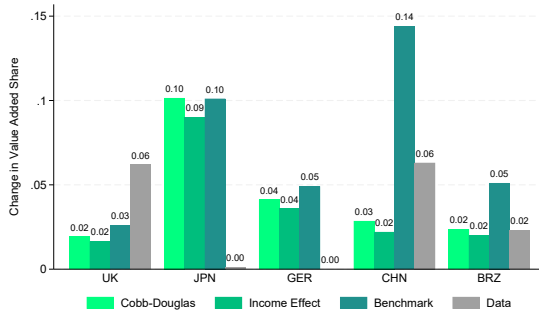
Table 4: Counterfactual: Dynamics of the United States

	Δ Prod. Gap	Δ VA. Share	Δ Emp. Share
Data	-12.2%	2.9%	4.9%
Benchmark	-4.4%	2.6%	2.8%
Income Effect	-3.1%	1.9%	2.0%
Population Aging	-3.8%	2.2%	2.4%

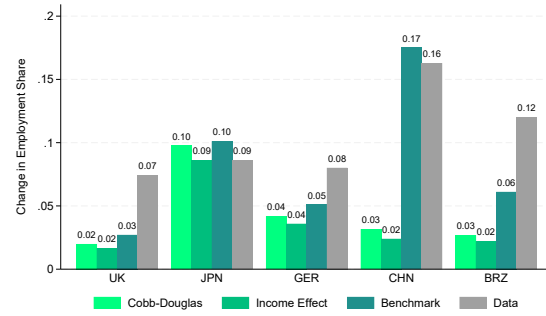
Notes: This table compares the model predicted dynamics of productivity gaps, value-added share of service sector and the employment share of sector from 2000 to 2017. The Benchmark row reports the model prediction in section 6.1. The Income Effect row reports a counterfactual with income growth effect only. The Population Aging row reports a counterfactual with Cobb-Douglas preference.



(a) Productivity Gaps



(b) Value-Added Share



(c) Employment Share

Figure 8: Counterfactual: Selected Countries

Notes: This graph compares the prediction of the model and the actual data on the dynamics of service-manufacturing Productivity Gaps, the Value-Added Share of service and the Employment Share of service. The light green bars in each panel are the predictions of the model when population aging is the only driver, while the dark green bars are the predictions of the model when both income effects and population aging are considered. The gray bars are the actual data for each selected countries. The selected countries are (in order) the United Kingdoms, Japan, Germany, China, Brazil.

6 Extended Model with Biased Technological Change

In the benchmark model, I showed that the decline in service sector productivity arises through the selection channel. Importantly, the benchmark model assumes that labor enters production symmetrically across sectors. This assumption abstracts from differences in labor and skill intensity between services and manufacturing, allowing the analysis to isolate the pure selection effects of labor reallocation driven by demand-side. In the extended model, I introduce skill-biased technical change (SBTC) production and sectoral capital-augmenting technical change (CATC) to

differentiate labor intensity and capital-augmenting efficiency of the service and manufacturing sectors. This extension allows me to analyze the selection effects from the production side, where more margin is reserved, where in the benchmark model, the changes in income and demographic structure are limited.

6.1 Labor Force

Compared to the benchmark, workers are now classified into high-skill and low-skill types to specify the contribution of different types of worker in the production. And each worker now have two components in their skill levels z_{kj}^i : general skill level κ_k and sector-specific skill. Similarly, the sector-specific skills are drawn from a joint Fréchet distribution $F(a_{ks}, a_{km}) = C[F(a_{ks}), F(a_{km})]$ governed by a Frank Copula.

$$F(a_{ks}) = e^{-a_{ks}^{-\theta_{ks}}} \quad \text{and} \quad F(a_{km}) = e^{-a_{km}^{-\theta_{km}}} \quad k = H, L.$$

Whether a worker has high or low general skills is decided by a binary distribution where the probability of worker to be high-skilled is π . Normalizing the general skill level of low-skill level worker equal to 1, we have

$$z_{Hj}^i = \kappa \cdot a_{Hj}^i, \quad z_{Lj}^i = a_{Lj}^i \quad j = s, m.$$

6.2 Environment

Production in both the service and manufacturing sectors follows a similar CES structure. Each sector includes a composite input that represents automatable tasks, which can be performed by either labor or capital. The production technologies for service and manufacturing are give by

$$Y_j = A_j \left[\alpha_j H_j^{\frac{\sigma-1}{\sigma}} + (1 - \alpha_j) X_j^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad X_j = \left[\mu_j L_j^{\frac{\xi-1}{\xi}} + (1 - \mu_j) (A_j^K K_j)^{\frac{\xi-1}{\xi}} \right]^{\frac{\xi}{\xi-1}} \quad j = s, m,$$

where Y_s is the output of service sector and Y_m is the output of manufacturing sector, H_j is the aggregate efficient labor of high-skilled worker and L_j is the aggregate efficient labor of low-skilled worker. X_j is the sector-specific production task composite, which is intuitively relative codifiable thus can be completed by both low-skilled workers and machine. A_j^K is referred to the capital-augmenting technical change. In the benchmark extension we follow Buera et al. (2022) and assume the elasticity of substitution between high-skilled labor and task composite σ is the same in both sectors; the elasticity of substitution between labor and capital ξ is also the same. And the efficient labor aggregators are given by

$$H_j = \int z_{Hj}^i dG(z_{Hs}^i, z_{Hm}^i), \quad L_j = \int z_{Lj}^i dG(z_{Ls}^i, z_{Lm}^i) \quad j = s, m.$$

6.3 Equilibrium

The representative firm in sector j solves the profit maximization problem:

$$\max_{H_j, K_j, L_j} \Pi_j = p_j A_j \left[\alpha_j H_j^{\frac{\sigma-1}{\sigma}} + (1 - \alpha_j) X_j^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} - w_H H_j - R_K K_j - w_L L_j$$

Given the first order conditions in the dual cost minimization problem in a competitive equilibrium, where the profits equal to zero, the price of sector j 's output can be expressed in terms of skill premium and capital premium

$$\hat{p}_j(w_H, R_K) = \frac{1}{A_j} \left\{ \frac{\alpha_j}{w_H^{\sigma-1}} + (1 - \alpha_j) \left[\mu_j + (1 - \mu_j) \left(\frac{R_K}{A_j^K} \right)^{1-\xi} \right]^{\frac{1-\sigma}{1-\xi}} \right\}^{\frac{1}{1-\sigma}}. \quad (20)$$

where the wage of low-skilled labor w_L is normalized to 1. Using the equilibrium conditions, we obtain the sector specific demand functions of high-skilled labor and capital in terms of w_H and R_K

$$H_j = \left[\frac{\alpha A_j \hat{p}_j(w_H, R_K)}{w_H} \right]^{\sigma} \frac{Y_j}{A_j}, \quad (21)$$

and

$$K_j = \left[\frac{(1 - \alpha_j) A_j \hat{p}_j(w_H, R_K)}{c_{jX}} \right]^{\sigma} \left[\frac{(1 - \mu_j) A_j^K c_{jX}}{R_K} \right]^{\xi} \frac{Y_j}{A_j A_j^K}, \quad (22)$$

where c_{jX} is the unit cost of composite production, which is given by the composite-specific cost minimization problem

$$c_{jX}(w_L, R_K) = \left[\mu_j^{\xi} w_L^{1-\xi} + (1 - \mu_j)^{\xi} \left(\frac{R_K}{A_j^K} \right)^{1-\xi} \right]^{\frac{1}{1-\xi}}.$$

Given the labor market equilibrium, workers choose the sector with highest pay and enter service sector if

$$\frac{z_{Hm}^i}{z_{Hs}^i} \leq \frac{\hat{p}_s}{\hat{p}_m} \quad \text{and} \quad \frac{z_{Lm}^i}{z_{Ls}^i} \leq \frac{\hat{p}_s}{\hat{p}_m}.$$

6.4 parameterization

Table 5: Esimated Parameters

Parameters	Definition	Value
π	Probability of being a HS worker	0.374
κ	General skill premium	1.166
θ_{Hs}	The dispersion level of HS worker's skill (service)	2.699
θ_{Hm}	The dispersion level of HS worker's skill (manufacturing)	3.588
θ_{Ls}	The dispersion level of LS worker's skill (service)	3.485
θ_{Lm}	The dispersion level of LS worker's skill (manufacturing)	4.325
ρ	The Frank Copula correlation	3.762
α_{s0}	Input share of HS worker in service at t=0	0.680
α_{sT}	Input share of HS worker in service at t=T	0.750
α_{m0}	Input share of HS worker in manufacturing at t=0	0.310
α_{mT}	Input share of HS worker in manufacturing at t=T	0.410
σ	Elasticity of substitution between HS labor and task composite	1.360
μ_s	Input share of LS labor in service	0.430
μ_m	Input share of LS labor in manufacturing	0.490
ξ	Elasticity of substitution between LS labor and capital	1.400
A_{mT}^K / A_{m0}	Accumulative change of capital-augmenting efficiency in manufacturing	[1.54, 2.71]

Notes: The details of the estimation process are shown in Online Appendix D.

Skill Distribution Using the same CPS dataset spanning from 2003 to 2018, we classified workers with college degree or above as High-Skill workers, and others are classified as Low-Skill workers⁵. Recalibrating the skill-distribution for High-Skill and Low-Skill workers, we obtain two sets of distribution parameters as well as skill premium and share of high-skilled labor.

Technology Parameters Start from the inner CES production, the compensation share of low-skilled labor is given by

$$\hat{s}_j^L = \frac{\mu_j^\xi w_{Lj}^{1-\xi}}{\mu_j^\xi w_{Lj}^{1-\xi} + (1 - \mu_j)^\xi (R_K / A_j^K)^{1-\xi}}. \quad (23)$$

Since we do not observe the user cost of capital at the sector level, we exploit the scale invariance of the CES aggregator and normalize the relative effective price of capital to low-skilled labor in the base year. I set $\left(\frac{R_{kt}}{w_{Ljt}}\right) = 1$ and recover subsequent changes in relative factor costs from observed movements in factor cost shares. I aggregate industry-level compensation data from BEA-KLEMS to generate the empirical share of low-skilled worker wage bill. I set the elasticity of substitution

⁵The classification in CPS follows: 1. Less Than High School (LTHS); 2. High School; 3. Some College (No degree) 4. College; 5. Advanced (Master or Above).

between low-skilled labor and capital $\xi = 1.40$ following the estimation of Hubmer (2023). Inverting equation 23 for a given ξ using different benchmark years, the implied values of the low-skilled input share range from 0.39 to 0.46 in services and from 0.45 to 0.53 in manufacturing with a declining trend. To estimate the elasticity of substitution between high-skilled labor and production composite, I aggregate the wage bill data from World-KLEMS to BEA-KLEMS to create an industry-sector panel from 1997 to 2010. Taking log to the input ratio equation, we obtain a estimation equation

$$\log \frac{\hat{s}_{it}^H}{1 - \hat{s}_{it}^H} = \gamma_i + \delta_t + (1 - \sigma) \log \frac{w_{H,s(i)t}}{c_{X,it}} + \epsilon_{it}, \quad (24)$$

where $\hat{s}_{it}^H$ is the industry compensation share from BEA-KLEMS, $w_{H,s(i)t}$ is the sector-level high-skilled wage constructed from Wolrd-KLEMS for the sector $s(i)$ to which industry i belongs, and $c_{X,it}$ is the unit cost of the task composite implied by the inner CES structure. And the proposed $\hat{\sigma}$ estimated from equation (24) is 1.40. Given $\hat{\sigma}$, inverting

$$\frac{\hat{s}_j^H}{1 - \hat{s}_j^H} = \left(\frac{\alpha_j}{1 - \alpha_j} \right)^{\hat{\sigma}} \left(\frac{\hat{w}_H}{\hat{c}_{X,j}} \right)^{1 - \hat{\sigma}} \quad (25)$$

and using 1997, 2005 and 2010 as benchmark years, the implied input share are in the ranges $\alpha_s \in [0.68, 0.75]$ and $\alpha_m \in [0.31, 0.41]$.

Capital-Augmenting Efficiency To calibrate the capital-augmenting efficiency overtime, I use the relative price of service compared to manufacturing as the target, and normalize the accumulative change of capital-augmenting efficiency of service sector $A_{sT}^K / A_{s0}^K = 1$. The first measure uses industry-level value-added price indices from the World KLEMS database (Jorgenson et al., 2012). Industries are aggregated into manufacturing and services using chained Fisher indices with nominal value-added weights, following standard national accounts procedures. This measure is closely aligned with the model concept of sectoral value-added prices but is only available through 2010. To extend the sample period, I construct a second measure using Consumer Price Index (CPI) data from the FRED. Specifically, I use the CPI for commodities and the CPI for services excluding rent of shelter, both for U.S. urban consumers. The relative price is defined as the ratio of the service CPI to the commodity CPI. While this measure is based on consumption prices rather than value-added prices, it provides consistent coverage through 2017. These two measures provide a broad interval for A_{mT}^K / A_{m0}^K , ranging from 1.54 to 2.70.

6.5 Quantitative Experiments

In this subsection, I use the calibrated model to quantitatively assess the changes in the productivity gap due to the exogenous SBTC and CATC. The quantitative results show that when skill-neutral

technical change is excluded, SBTC and CATC can generate opposite effects under general equilibrium. Table 6 presents the results of the set of quantitative experiments.

SBTC and CATC When skill-neutral technical change and CATC are excluded, SBTC generates positive selection into services: the average skill/talent of workers in services rises over time even. In our parameterization, the intensity of high-skilled labor increases more in the manufacturing sector compared to the service sector, which aligns with the conclusion of Buera et al. (2022).

In Column (i), the economy features SBTC but no manufacturing CATC. In the characterized equilibrium, the model predicts an increase in the service–manufacturing productivity gap. The key driver is a composition effect with the opposite sign relative to the demand-driven benchmark. Because SBTC is stronger in manufacturing, it raises the relative return to skill in manufacturing and shifts the equilibrium wage structure in a way that makes entering the service sector more selective. With capital deepening not increasing in services, the rise in relative service productivity is primarily driven by this positive selection mechanism induced by production-side skill bias, rather than by capital accumulation in services.

Columns (ii) to (iv) display the quantitative results when CATC is included in the model. Column (ii) shows that even when the selection mechanism decreases the average skill in the service sector, there could still be a productivity gain for the service sector. This result shows that the increase in relative price in services has two opposite effects on the relative productivity gap between the service sector and the manufacturing sector: the selection effects and the capital-labor substitution effect.

The selection effects decrease the productivity of the service sector by introducing less productive workers, when $\Delta A_{mt}^K = 0.5$, the average skill level in the service sector decreases 1.30% and increases 1.28% for the manufacturing sector. But the relative productivity of the service sector still increases due to the existence of capital-labor substitution effect. Higher $\mathbf{p}_s$ increase the marginal production in capital of the service sector, so the service sector will rent more capital, which offsets the productivity loss generated by selection effects.

Columns (iii) and (iv) show that when the CATC is substantial enough, the selection effects will dominate the capital-labor substitution effects. When $\Delta A_{mt}^K = 1$, the predicted productivity gap change is 4.58% while the average skill gap is 7.04%, which shows that the capital-labor substitution effects partially offset the productivity loss. When $\Delta A_{mt}^K = 1.7$, the size of the productivity loss is approximately equal to the difference in the change in the average skill level across the two sectors, since the marginal production in capital is decreasing.

Table 6: Extended Model: SBTC and CATC

	(i) $\Delta A_{mt}^K = 0$	(ii) $\Delta A_{mt}^K = 0.5$	(iii) $\Delta A_{mt}^K = 1$	(iv) $\Delta A_{mt}^K = 1.7$
$\Delta \text{Prod. Gap}$	+10.30%	+1.83%	-4.58%	-10.76%
$\Delta \text{Ave. Skill in Serv.}$	+1.45%	-1.30%	-2.84%	-3.74%
$\Delta \text{Ave. Skill in Manu.}$	-1.10%	+1.28%	+4.20%	+7.85%
$\Delta K_{s,t}$	-1.48%	+11.39%	+30.23%	+63.76%

Notes: This table reports model-implied percentage changes in the productivity gap, average skill levels in the service and manufacturing sectors, and capital allocation to the service sector for the United States. Column (i) presents the model with skill-biased technical change (SBTC) only, while Columns (ii)–(iv) additionally incorporate capital-augmenting technical change (CATC) in the manufacturing sector, with $A_{mT}^K/A_{m0}^K = \{1.5, 2, 2.7\}$. The productivity gap and average skill levels are reported as absolute differences.

Decomposing Selection Effects In Table 6, we show that the selection effects will be partially offset by the capital-labor substitution effects. We now decompose the selection effects and capital-labor substitution effects. The results are presented in Table 7.

In Panel A, both the elasticity of substitutions is set to 1, and the production now follows a Cobb-Douglas technology where labor and capital are close to perfect complements. When there are no capital-labor substitution effects, the predicted productivity gap changes and the difference in the change of sectoral average skill level are modestly higher than the nested CES case. When $\Delta K_{m,t}^K = 1.7$, the change in productivity gaps shifts from -10.76% to -11.67%. This result arises because the primary driver of structural change in the model is not substitution in production, but selection effects from labor reallocation. Even under Cobb-Douglas production, where $Y_j = A_j H_j^{\alpha_j} L_i^{(1-\alpha_j)\mu_j} (A_j^K K_j)^{(1-\alpha_j)(1-\mu_j)}$, marginal products depend on input factor quantities, so changes in the relative supply of skill across sectors continue to generate wage differentials and induce worker reallocation.

In Panel B, only the elasticity of substitution between high-skilled labor and task composite is set to 1, and now these two input factors are complements. The results show that the selection effects are amplified when capital only substitutes low-skilled labor. When CATC raises the capital-augmenting efficiency in the manufacturing sector, low-skilled workers will be displaced, increasing the inflow of low-skilled labor to the service sector. Since low-skilled labor and high-skilled labor are non-substitutable now, this will amplify the negative selection effect and leads to a larger decline in relative service productivity.

In contrast, when only the substitution between capital and high-skilled labor is allowed in Panel C, capital replaces high-skilled workers in the manufacturing sector, increasing the relative supply of high-skilled workers in the service sector. This mitigates the negative selection effect and dampens the decline in relative service productivity significantly. For each level of CATC considered, Panel C exhibits the smallest change in the average skill level in the service sector

relative to Panels A and B as well as the benchmark nested CES in Table 6.

Table 7: Extended Model: Role of Substitution in Selection Effects

	(i) $\Delta A_{mt}^K = 0$	(ii) $\Delta A_{mt}^K = 0.5$	(iii) $\Delta A_{mt}^K = 1$	(iv) $\Delta A_{mt}^K = 1.7$
<i>Panel A. Cobb-Douglas ($\sigma = 1, \xi = 1$)</i>				
$\Delta \text{Prod. Gap}$	+11.25%	+0.96%	-5.67%	-11.67%
$\Delta \text{Ave. Skill in Serv.}$	+1.31%	-1.88%	-3.53%	-4.66%
$\Delta \text{Ave. Skill in Manu.}$	-0.52%	+1.49%	+3.34%	+5.51%
$\Delta K_{s,t}$	-1.02%	+17.15%	+33.38%	+51.95%
<i>Panel B. Capital-low-skill substitution only ($\sigma = 1, \xi = 1.40$)</i>				
$\Delta \text{Prod. Gap}$	+11.35%	+0.00%	-8.26%	-16.13%
$\Delta \text{Ave. Skill in Serv.}$	+1.34%	-2.14%	-4.09%	-5.29%
$\Delta \text{Ave. Skill in Manu.}$	-0.51%	+1.73%	+4.41%	+7.70%
$\Delta K_{s,t}$	-1.38%	+13.43%	+31.07%	+56.34%
<i>Panel C. Capital-High-skill substitution only ($\sigma = 1.36, \xi = 1$)</i>				
$\Delta \text{Prod. Gap}$	+10.30%	+2.41%	-2.97%	-7.72%
$\Delta \text{Ave. Skill in Serv.}$	+1.45%	-1.11%	-2.51%	-3.38%
$\Delta \text{Ave. Skill in Manu.}$	-1.13%	+1.02%	+3.52%	+6.16%
$\Delta K_{s,t}$	-1.13%	+16.97%	+34.87%	+58.94%

Notes: This table reports model-implied changes between the initial and terminal periods. The productivity gap and average skill levels are reported as absolute differences. The change in capital allocated to services, $\Delta K_{s,t}$, is reported as the percentage change relative to its initial level. Panel A sets $\sigma = \xi = 1$. Panel B sets $\sigma = 1$ and $\xi = 1.40$. Panel C sets $\sigma = 1.36$ and $\xi = 1$. Columns (ii)–(iv) incorporate capital-augmenting technical change in manufacturing.

Decomposing Capital-Labor Substitution Effects Table 8 displays the result of the change in productivity gap and the percentage change in capital rented by the service sector when the selection channel is eliminated. To isolate the role of capital-labor substitution from compositional selection effects, I construct a counterfactual economy in which the labor allocation across sectors is fixed at its initial equilibrium configuration. As a result, labor reallocation and compositional changes in sector-specific efficiency are shut down, while capital reallocation, capital-labor substitution, and general equilibrium price adjustments remain fully operative.

In Column (i), where only SBTC is active and no manufacturing CATC is introduced, the productivity gap increases. The faster SBTC in the manufacturing sector decrease the relative price of services. This reflects a general-equilibrium supply response: with labor allocation fixed, SBTC and capital-labor substitution reshape sectoral unit costs and induce capital deepening in services, raising service labor productivity relative to manufacturing.

Once manufacturing CATC is introduced (Columns ii–iv), manufacturing becomes increasingly

attractive for capital, which shifts capital away from services. As the strength of CATC rises, service capital deepening weakens and the productivity gap eventually declines. The direction of the productivity gap closely tracks the direction of capital reallocation. This contrasts sharply with the baseline economy with endogenous selection. In that case, even when service capital deepening increases, the productivity gap can still decline because compositional effects dominate. The entry of marginal workers into the expanding sector reduces average efficiency and offsets the positive effect of capital accumulation. The comparison demonstrates that selection effects can overturn the effects of capital deepening and reverse the direction of relative productivity gap movements.

Table 8: Extended Model: Decomposing Substitution Effects

	(i) $\Delta A_{mt}^K = 0$	(ii) $\Delta A_{mt}^K = 0.5$	(iii) $\Delta A_{mt}^K = 1$	(iv) $\Delta A_{mt}^K = 1.7$
$\Delta \text{Prod. Gap}$	+9.05%	+3.37%	-0.18%	-3.48%
$\Delta K_{s,t}$	+8.9%	-2.10%	-9.48%	-16.79%

Notes: This table reports model-implied percentage changes in the productivity gap and capital allocation to the service sector for the United States. The productivity gap is reported as absolute differences. The change in capital allocated to services, $\Delta K_{s,t}$, is reported as the percentage change relative to its initial level. Columns (ii)–(iv) incorporate capital-augmenting technical change in manufacturing.

7 Conclusion

This paper shows that population aging can have significant implications for sectoral productivity through a demand-driven selection mechanism. As retirees exhibit stronger preferences for services, aging increases the relative price of services and expands the service sector. In a Roy framework, sectoral expansion draws in marginal workers with weaker comparative advantage, lowering measured service-sector productivity through composition effects.

Calibrated to U.S. micro-data, the benchmark model accounts for 36% of the decline in the U.S. service–manufacturing productivity gap and replicates the dynamics of service employment shares across high-income countries. Although demographic forces alone cannot fully explain cross-country differences—especially in developing economies with more pronounced technological heterogeneity stages—the model performs particularly well in matching sectoral labor reallocation patterns. This indicates that the selection mechanism is quantitatively important for understanding how demographic aging and income-driven demand shifts reshape labor allocation and structural transformation.

Extending the model to include skill-biased technical change and capital-augmenting technical change introduces additional production-side margins that improve the quantitative fit of productivity dynamics. Quantitatively, production-side technological change is also essential

for explaining movements in relative service productivity. In particular, a faster skill-biased technical change in manufacturing induces sectoral expansion in manufacturing, which lowers average manufacturing productivity under the Roy framework. When capital-augmenting technical change is sufficiently strong, the resulting increase in relative prices raises the selection threshold, and selection effects then reduce service-sector productivity. Further decomposing selection effects and capital–labor substitution effects, I find that although biased technological change and capital deepening can exert opposing influences on sectoral productivity, the selection channel remains quantitatively important in explaining the evolution of the U.S. service–manufacturing productivity gap.

Overall, the results suggest that population aging is not only a driver of structural transformation, but also a force shaping measured productivity differences across sectors. As demographic transitions continue worldwide, understanding the compositional effects embedded in service-sector expansion becomes increasingly important for interpreting long-run productivity trends.

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